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Earned Value Management Exercises

Based on the PMBOK® Guide Sixth Edition

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Practice Exercises

Earned Value Management

Exercise 1:

You're a subcontractor responsible for refurbishment of an automobile showroom. The estimated refurbishment cost is $500 per square feet. The total showroom area that needs to be refurbished is 1,000 square feet. Based on your past experience, you know your team can refurbish 100 square feet per week.

After 4 weeks, you have 45% of the job complete and you have spent $250,000.

Determine the value for each of the terms below:

Term	Value
Budget At Completion	
Planned Value	
Earned Value	
Actual Cost	
Cost Variance	
Schedule Variance	
Cost Performance Index	
Schedule Performance Index	
Estimate At Completion	
Estimate To Complete	
Variance At Completion	

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Exercise 2:

Your project is required to install lighting fixtures in a 200 room hotel. You have to install 5 lighting fixtures in each room. Your project's approved budget is $500,000 and the approved schedule is 14 weeks.

After 8 weeks, you have 110 of the rooms completed and you have spent $250,000.

Determine the value for each of the terms below:

Term	Value
Budget At Completion	
Planned Value	
Earned Value	
Actual Cost	
Cost Variance	
Schedule Variance	
Cost Performance Index	
Schedule Performance Index	
Estimate At Completion	
Estimate To Complete	
Variance At Completion	

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Exercise 3:

Your project is tasked with excavation of 1,000,000 cubic feet of soil. According to the project management plan you need to excavate 60,000 cubic feet of soil per week at a cost of $5 per cubic feet. According to the first month's progress report the project's EV was 1,000,000 and the AC was 1,200,000. According to the second month's progress report the project's EV was 2,200,000 and the AC was 4,400,000. After the 3rd month (12 weeks) you have 250,000 cubic feet of soil remaining to be excavated and you total cost to date is 4,000,000.

Determine the current values for the project for each of the terms below:

Term	Value
Budget At Completion	
Planned Value	
Earned Value	
Actual Cost	
Cost Variance	
Schedule Variance	
Cost Performance Index	
Schedule Performance Index	
Estimate At Completion	
Estimate To Complete	
Variance At Completion	

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Exercise 4:

You are the Program Manager managing the construction of a new airport. Five project managers are directly reporting to you. Each project manager is assigned a part of the total scope of works to be executed independently. The total duration allowed to complete all works is 24 months and the total approved budget is $10,000,000.

At the end of the first year, you should have completed $6,000,000 worth of work but the project is only 50% complete. Actual costs reported by each project manager for the first year are:

Project manager 1: $200,000

Project manager 2: $1,200,000

Project manager 3: $1,800,000

Project manager 4: $500,000

Project manager 5: $900,000

Determine the current values for the entire program for each of the terms below:

Term	Value
Budget At Completion	
Planned Value	
Earned Value	
Actual Cost	
Cost Variance	
Schedule Variance	
Cost Performance Index	
Schedule Performance Index	
Estimate At Completion	
Estimate To Complete	
Variance At Completion	

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Earned Value Management Exercise
Solutions

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Exercise 1 Solution:

Estimated refurbishment cost	500.00	per square feet
Total showroom area	1,000.00	square feet
Refurbishment pace	100.00	square feet per week
Budget At Completion ($500 x 1,000)	**500,000.00**	**dollars**
Total duration (1,000 / 100)	10.00	weeks
Duration Elapsed	4.00	weeks
Planned Value (4 / 10 x BAC)	**200,000.00**	**dollars**
Percentage completion	45%	
Earned Value (45% x BAC)	**225,000.00**	**dollars**
Actual Cost	**250,000.00**	**dollars**
Cost Variance (EV - AC)	**(25,000.00)**	**dollars**
Schedule Variance (EV - PV)	**25,000.00**	**dollars**
Cost Performance Index (EV / AC)	**0.90**	
Schedule Performance Index (EV / PV)	**1.13**	
Estimate At Completion (BAC / CPI)	**555,555.56**	**dollars**
Estimate To Complete (EAC - AC)	**305,555.56**	**dollars**
Variance At Completion (BAC - EAC)	**(55,555.56)**	**dollars**

Exercise 2 Solution:

Budget At Completion	**500,000.00**	**dollars**
Approved duration	14.00	weeks
Duration Elapsed	8.00	weeks
Planned Value (8 / 14 x BAC)	**285,714.29**	**dollars**
Completed scope of works	110.00	rooms
Total scope of works	200.00	rooms
Percentage completion (110/200)	55%	
Earned Value (55% x BAC)	**275,000.00**	**dollars**
Actual Cost	**250,000.00**	**dollars**
Cost Variance (EV - AC)	**25,000.00**	**dollars**
Schedule Variance (EV - PV)	**(10,714.29)**	**dollars**
Cost Performance Index (EV / AC)	**1.10**	
Schedule Performance Index (EV / PV)	**0.96**	
Estimate At Completion (BAC / CPI)	**454,545.45**	**dollars**
Estimate To Complete (EAC - AC)	**204,545.45**	**dollars**
Variance At Completion (BAC - EAC)	**45,454.55**	**dollars**

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Exercise 3 Solution:

Total scope of works	1,000,000.00	cubic feet
Remaining scope of works	250,000.00	cubic feet
Completed scope of works (Total - Remaining)	750,000.00	cubic feet
Percent completion	75.00%	
Weekly target	60,000.00	cubic feet per week
Total duration (SOW / target)	16.6667	weeks
Budgeted cost per week (target x $5)	300,000.00	dollars per week
Budget At Completion (cost per week x duration)	**5,000,000.00**	**dollars**
Duration Elapsed	12.00	weeks
Planned Value (12 / 16.6667 x BAC)	**3,600,000.00**	**dollars**
Earned Value (Percent complete x BAC)	**3,750,000.00**	**dollars**
Actual Cost	**4,000,000.00**	**dollars**
Cost Variance (EV - AC)	**(250,000.00)**	**dollars**
Schedule Variance (EV - PV)	**150,000.00**	**dollars**
Cost Performance Index (EV / AC)	**0.94**	
Schedule Performance Index (EV / PV)	**1.04**	
Estimate At Completion (BAC / CPI)	**5,333,333.33**	**dollars**
Estimate To Complete (EAC - AC)	**1,333,333.33**	**dollars**
Variance At Completion (BAC - EAC)	**(333,333.33)**	**dollars**

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Exercise 4 Solution:

Budget At Completion	**10,000,000.00**	**dollars**
Planned Value	**6,000,000.00**	**dollars**
Percent complete	50%	
Earned Value (Percent complete x BAC)	**5,000,000.00**	**dollars**
AC - Project 1	$200,000	dollars
AC - Project 2	$1,200,000	dollars
AC - Project 3	$1,800,000	dollars
AC - Project 4	$500,000	dollars
AC - Project 5	$900,000	dollars
Total Actual Cost	**$4,600,000**	**dollars**
Cost Variance (EV - AC)	**400,000.00**	**dollars**
Schedule Variance (EV - PV)	**(1,000,000.00)**	**dollars**
Cost Performance Index (EV / AC)	**1.09**	
Schedule Performance Index (EV / PV)	**0.83**	
Estimate At Completion (BAC / CPI)	**9,200,000.00**	**dollars**
Estimate To Complete (EAC - AC)	**4,600,000.00**	**dollars**
Variance At Completion (BAC - EAC)	**800,000.00**	**dollars**

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